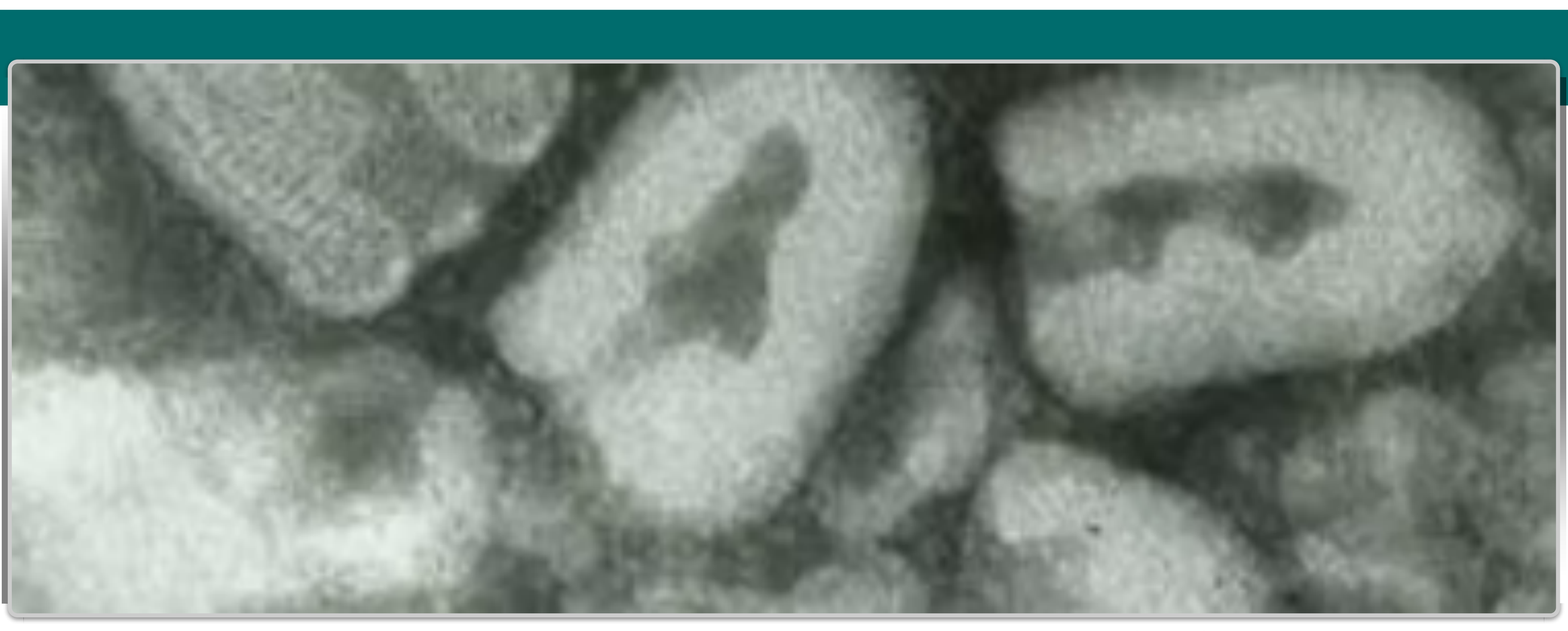

Photo credit: Dr.Geert Wiegertjes, Wageningen University

Return to Gallery of Invaders (Pathogens)

Spring Viremia of Carp (SVC)

(Rhabdovirus carpio)

Report this Species!

If you believe you have found this species anywhere in Pennsylvania, please report your findings to state and federal authorities immediately!

- **Pennsylvania Fish & Boat Commission** (state authority)
- **U.S. Fish & Wildlife Service** (federal authority)

Species at a Glance

The spring viremia of carp (SVC) virus is a highly contagious pathogen found to impact mainly **common carp** and related species. While fish can carry SVC with or without symptoms, outbreaks can cause high rates of death and substantial economic losses.

Note: As of September 2016, the U.S.Geological Survey has detected that SVC can be transmitted to amphibians (in addition to fish). Read the **full news story** and **original published study**.

Identification

SVC is caused by a bullet-shaped RNA virus that can cause external symptoms including bulging eyes; hemorrhaging of the skin, gills, and eyes; a bloated appearance; darkening of the skin; and vent protrusion. Internally, fluids can build up in the organs and body cavity, and hemorrhaging and inflammation can occur in the swim bladder and intestines. Diseased fish may appear lethargic, swim and breathe more slowly than normal, and tend to gather at the water inlet or sides of ponds. Loss of equilibrium with resting and leaning can also be seen in later stages. Other members of the Family Cyprinidae (minnow family) and possibly northern pike (*Esox lucius*) are also susceptible to this disease.

Similar Species

The symptoms of SVC have characteristics similar to those of other fish diseases, so lab testing is necessary to confirm that a fish is infected with SVC.

Habitat

SVC usually occurs in the spring, when water temperatures are less than 18°C (64°F); however, the virus can persist in 10°C (50° F) water for more than four weeks, and in 4° C (39° F) mud for at least six weeks. Mortality rates vary with stress factors, population density, age, and condition of the fish.

Spread

SVC is highly contagious and can be spread through contact with the environment and through parasitic invertebrates such as the carp louse or leeches. Infected fish shed the virus through feces, urine, and gill and skin mucous. When fish come into contact with infected water, the virus enters most often through the gills.

Distribution

SVC has been identified in Europe, Russia, and the Middle East, and was first diagnosed in Yugoslavia. The first report of this disease in the United States was in 2002 in North Carolina. It has since been found in Wisconsin, Illinois, Missouri, Washington, Lake Michigan, and in areas of the upper Mississippi River.

Note: Distribution data for this species may have changed since the publication of ***Pennsylvania's Field Guide to Aquatic Invasive Species (Second Edition 2015)****, the source of information for this description.*

Environmental Impacts

SVC mortality rates, which can reach up to 70 percent in young carp, can have serious economic impacts on aquaculture. No vaccines exist for the virus, so prevention by proper disinfection of equipment and gear is the best method for fighting this disease. There is currently no indication that SVC is any threat to human health.

Note

Information for this species profile comes from **Pennsylvania's Field Guide to Aquatic Invasive Species (Second Edition 2015)**.

The Pennsylvania iMapInvasives Program is a partnership of the Western Pennsylvania Conservancy, the Pennsylvania Natural Heritage Program, and NatureServe.
Funding for Pennsylvania iMapInvasives is provided by the Great Lakes Restoration Initiative.